



NIKE VELOCITY

Technology Assignment – Group 2

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Team Introduction



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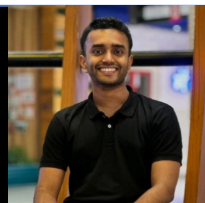
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Agenda



01

From Insight to Innovation

Synthesizing former analyses to identify key areas where AI can drive the most impact for Nike

02

AI-Powered Operations & Supply Chain

AI solutions to optimize Nike's demand forecasting, inventory management, and global logistics efficiency

03

Customer Experience

Reimagining how customers discover and buy Velocity products through AI-driven innovations

04

Data & AI Intelligence

Mapping the underlying AI models, data requirements, and system integrations needed to bring Velocity to life

05

Challenges & Next Steps

Key implementation barriers and a prioritized roadmap for scaling Nike's AI transformation forward

06

Appendix

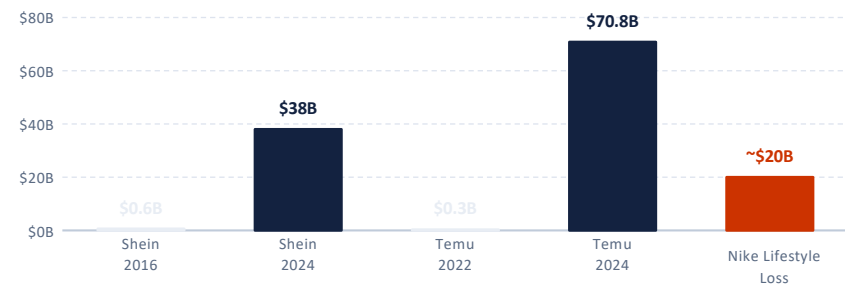
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From Insight to Innovation

Velocity enters fast fashion as a Nike subsidiary — brand, data, and members are the moat no fast-fashion rival can replicate



FAST-FASHION MARKET NIKE IS MISSING · GMV/Revenue (USD B)



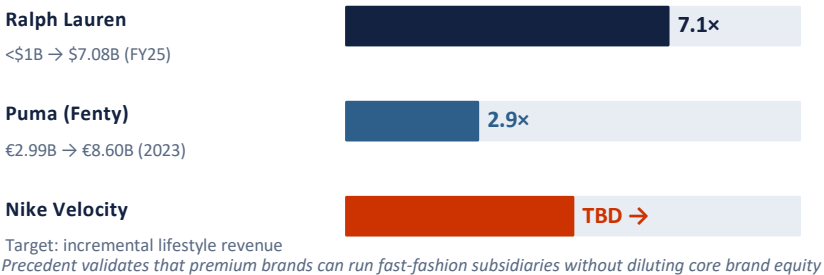
Shein 62x growth · Temu 244x growth · Nike lifestyle segment (40–50% of revenue) in decline

VELOCITY'S UNFAIR ADVANTAGE · What fast-fashion rivals cannot replicate



Shein & Temu have speed and AI — but zero brand equity, zero member data, zero cultural credibility

STRUCTURAL PRECEDENT · Sub-brand revenue growth multiple



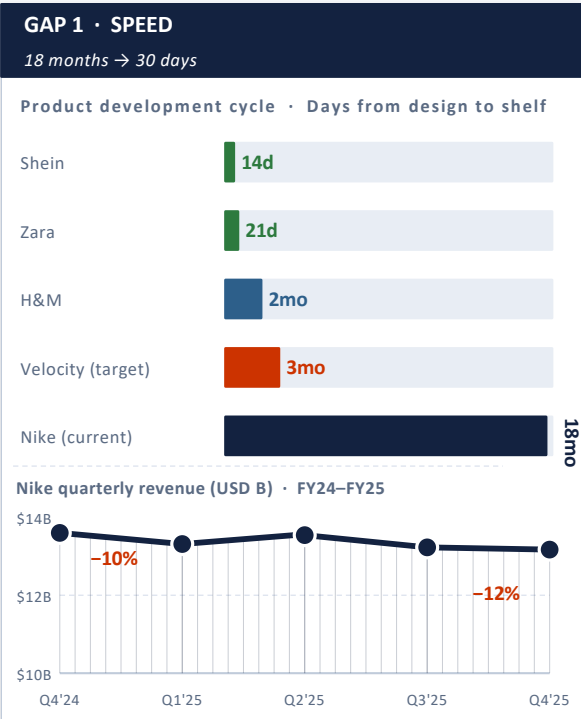
Target: incremental lifestyle revenue
Precedent validates that premium brands can run fast-fashion subsidiaries without diluting core brand equity

WHY VELOCITY WORKS AS A SUBSIDIARY · Key inferences

Market	Fast fashion is validated and growing. Shein+Temu = \$108B combined GMV. Gen Z already spends \$2,388/yr on apparel. Velocity enters an existing, sized market.
Architecture	Ralph Lauren (7x) and Puma (2.9x) prove the sub-brand model works. Velocity operates independently — Nike's Swoosh provides launch credibility, not operational constraints.
Consumer	Gen Z already segments spend: Nike for hero purchases (\$120–200+), fast-fashion for weekly trends (\$20–60). Velocity fits the lower tier without cannibalizing Jordan/Dunk.
Moat	Speed alone is not the differentiator — Shein has that. Velocity's moat is speed + Nike's 160M member demand intelligence. Structurally unreplicable by any pure-play rival.

Both market research and user feedback suggest the same solution: make Velocity a separate fast-fashion brand under Nike. That way, it can operate with speed and flexibility while still using Nike's brand power and insights, giving it an edge others cannot copy.

Three capability gaps separate Velocity from fast-fashion competitiveness — Nike's AI infrastructure closes each one



Nike has: Celect + Zodiac + A.I.R. — connecting them into a 30-day pipeline is the build question for the tech deck.



Nike's 160M member + athlete network = upstream trend signals. Can this be operationalized as a real-time brand-code filter?

GAP 3 · CHANNEL

Nike.com → Social-native

The TikTok discovery-to-purchase loop · Nike vs. Velocity

STEP	NIKE TODAY	VELOCITY
1 Trend surfaces	Manual / seasonal	AI detection (real-time)
2 Design to shelf	12–18 months	Target: <30 days
3 Channel	Nike.com (DTC)	TikTok Shop + IG native
4 Response	Cannot respond fast	Drop-ready in window
5 Feedback loop	None / slow	72hr sell-through → AI

48% of 18–34 yr-olds make fashion purchases via social — Nike's catalog UX is fundamentally wrong for this consumer

Gen Z AI readiness · Velocity model alignment

Trend detection / social drops	HIGH ✓
Small batch / scarcity mechanics	HIGH ✓
AI-curated product feed	MED ✓

Three gaps, three AI interventions, three tech questions — the technology assessment must now determine whether the required stack is build, buy, or partner, and at what speed Nike can

From Insight to AI Opportunity



01 THREE KEY GAPS

1 Product Speed

FROM

18-Month Cycle

→ 30-Day Drops

2 Relevance

FROM

Seasonal Collections

→ Culturally Current

3 Channel

FROM

Nike.com

→ Social-Native Commerce

02 AI SOLUTION AREAS

Former analyses point to two critical AI intervention areas:



Operations & Supply Chain

Speed up the make cycle — AI-driven demand forecasting, agile manufacturing, and real-time inventory optimization.



Data & System Integration

Power real-time cultural relevance — unified data pipelines, trend intelligence, and social-commerce AI layers.

03 CUSTOMER EXPERIENCE SHIFT

Experience	Old Nike	Nike Velocity AI
Discovery	Browse Nike.com	Served via social feed
Product Cycle	Wait months	Drop within 30 days
Relevance	Seasonal catalog	Culturally triggered
Purchase	Website checkout	Social-native instant buy

Former analyses converge on a single diagnosis: Nike is losing on speed, not quality.

AI-Powered Operations & Supply Chain

AI Solution: Real-Time Demand Sensing



Bottleneck 1: The demand signal arrives too late

Nike validates demand through retail sell-through data, feedback that arrives months after production is already locked. By the time a trend is confirmed internally, the design-to-production clock has already burned 6 to 9 months. Velocity cannot afford to wait that long.

Solution: Real-Time Demand Sensing

PRODUCTION-READY

What we recommend

Nike licenses WGSN Instock or Trendalytics and connects it to the Nike data lake. Social trend velocity from TikTok, Instagram, and Pinterest is layered with 160M member purchase signals to produce a weekly ranked list of Velocity drop candidates, each with a confidence score.

How the output works

High confidence score triggers a full production order. Low confidence triggers a small test batch. The system decides, not a planner. This directly eliminates the guesswork driving Nike's current overproduction and markdown cycles.

Why this is Velocity's competitive moat

Shein reads public trend data. Velocity reads public trend data filtered through Nike's own 160M members. That is the difference between fast and Nike-specific fast. No competitor can replicate this signal.

Expected Impact

20–30%

reduction in unsold inventory, based on comparable deployments at Zara and ASOS

Recommended platforms

- **WGSN Instock**

Scans social media to detect emerging trends 2 to 4 weeks before they peak

- **Trendalytics**

Translates trend signals into ranked drop candidates with confidence scores

Already proven by

Zara | ASOS | Stitch Fix

Nike licenses proven infrastructure and trains it on proprietary NikePlus data. No build from scratch required.

Nike already has 160M members. Velocity gives that data a job: predicting exactly what to make before production begins.

AI Solution 2: AI-Driven Micro-Batch Production



Bottleneck 2: Production has no mechanism to respond in real time

Supplier contracts are fixed seasons in advance. If a colorway spikes mid-drop, Nike has no way to reallocate manufacturing capacity without restarting a new planning cycle. Every major fast fashion competitor has solved this. Nike has not.

Solution: AI-Driven Micro-Batch Production

PRODUCTION-READY (Platform)
/ EMERGING (Nike Integration)

What we recommend

Nike establishes a pre-qualified nearshore manufacturing network (2 to 3 partners in Mexico or Portugal) integrated via a shared production orchestration platform: o9 Solutions or Kinaxis, both already proven in apparel. When a demand signal clears the confidence threshold, a production order fires automatically. No manual planning cycle required.

How batch size is decided

The confidence score from Solution 1 sets the batch size automatically. If a specific colorway spikes mid-drop, the system re-routes production capacity across the nearshore network in real time, without waiting for the next planning cycle.

Why nearshore specifically

Nearshore partners in Mexico and Portugal offer 2 to 3 week turnaround vs. 8 to 12 weeks from traditional Asian suppliers. This is where Velocity's 30-day target becomes physically possible.

Combined Impact of Solutions 1 + 2

Nike's current design-to-shelf cycle

12 to 18 months

↓ toward 30 days

Velocity's target, on par with Zara, at Nike's quality standard

Recommended platforms

- **o9 Solutions**
Connects demand signals to production scheduling in real time
- **Kinaxis**
Automates production re-routing across manufacturer networks

Already proven in apparel by

Inditex (Zara's parent) | H&M Group

Demand Sensing tells Velocity what to make. Micro-Batch Production makes it happen in days, not months.

Data & AI Intelligence

The Underlying AI Technologies



Four AI systems powering Velocity's intelligence layer

Social Trend Forecasting

NLP (Language) + Computer Vision (Image Recognition)

Scans TikTok, Instagram & Pinterest to detect trends 2–4 weeks before they peak

● Available Now

*Commercially deployed across fashion and retail
Multiple vendors offer this as a plug-in service today*

Generative AI Design Studio

Diffusion Models / GANs (Image Generation AI)

Auto-generates colorways & designs from trend signals — hours, not weeks

● In Development

Actively piloted by early adopters like Adidas and Adobe, but not yet standardized at enterprise fashion scale

Drop Performance Prediction

ML Classifier / Predictive Analytics (AI Models)

Predicts sell-through rate & hype score before committing to production — Produce or Reject

● Available Now

Proven by Stitch Fix & ASOS — predictive scoring is standard retail technology today

Unified Customer Data Platform

Data Lake + Identity Resolution (Single Customer Profile System)

Aggregates all Nike apps & in-store data into one profile — powers all AI tools

● Available Now

Industry standard infrastructure — fine-tuned and synced in real time for Velocity's weekly drop cycle

Trend Forecasting spots it. Generative AI creates it. Drop Prediction approves it. Unified Data powers it all.

Data Requirements



INTERNAL DATA

What Nike Already Has

Existing Nike membership data

Purchase history, loyalty behavior

App browsing & engagement

What users view, save, skip

Past drop performance

Which designs sold, flopped, returned

In-store behavior

Foot traffic, try-on, purchase patterns

Inventory & production records

Historical stock and fulfillment data

+

EXTERNAL DATA

What Velocity Plugs Into

TikTok, Instagram, Pinterest

Real-time trend signals & aesthetics on social media

StockX & Depop resale platforms

Resale demand & hype indicators

Professional trend forecasting services

Enterprise trend forecasting feeds, eg. WGSN, Trendalytics or Heuritech

Search engine trend data

Google Trends, keyword volume signals

Competitor pricing data

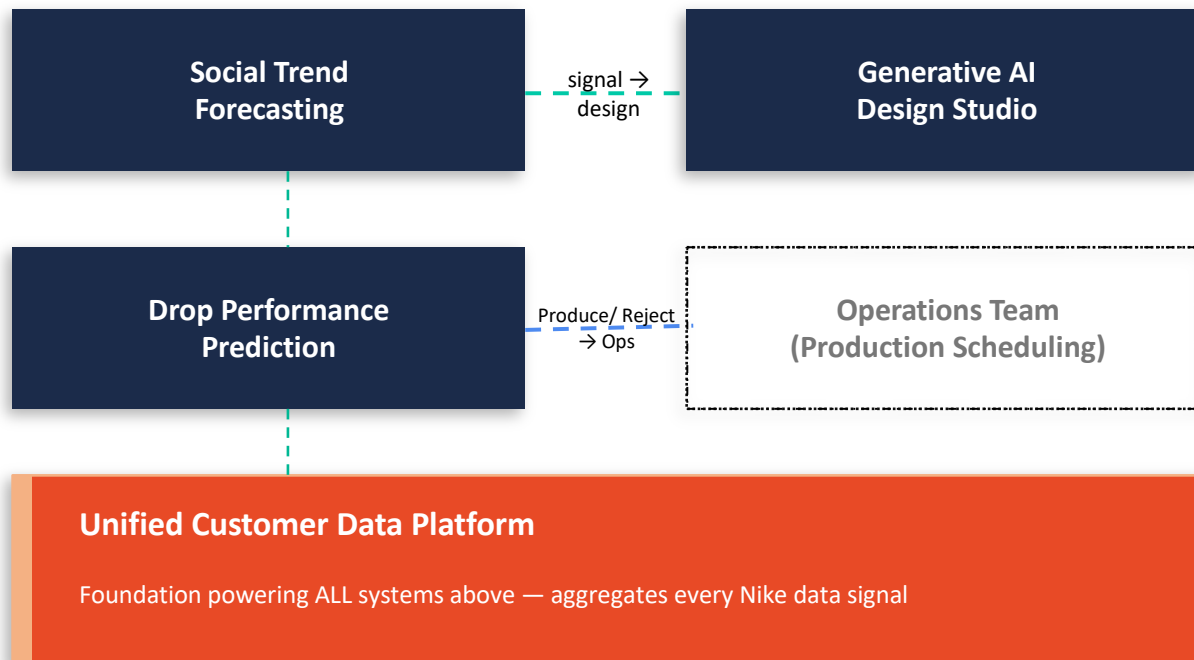
Market positioning & price benchmarks

Data feeds the AI systems — internal signals and external intelligence

System Integration & Dependencies



How the four AI systems connect



KEY CHALLENGES

- **ERP Integration**
Connecting Velocity AI to Nike's existing supply chain & inventory systems
- **Data Privacy Law**
GDPR (EU) & CCPA (US) compliance across all customer data collection points
- **Design Adoption**
Getting design teams to trust and work with AI-generated outputs
- **Real-time Sync**
Keeping the Unified Data Platform updated instantly across all inputs

Four systems, one language — the intelligence layer is what makes fast fashion actually intelligent

Customer Experience

Velocity's AI transforms every stage of the drop journey, from how customers discover a drop to what they do after buying



THE VELOCITY CUSTOMER EXPERIENCE • FOUR FRONTIERS

CX Frontier	Where Nike Falls Short	How Velocity Closes It	CX Outcome
Drop Discovery	Generic push notifications and email blasts. Every member receives the same announcement regardless of cultural relevance.	AI surfaces each Velocity drop through TikTok-native social channels, reaching the right cultural cohort where they already spend time.	Discovery feels like an algorithmic recommendation, not a mass marketing blast. Zero friction between seeing the drop and acting on it.
Drop Access	SNKRS lottery randomness. Bots win systematically. Real fans lose to automated scripts and disengage permanently.	Reinforcement learning model scores waitlist members by engagement authenticity, filtering bots and prioritising genuine community members.	Losing a drop feels fair. Trust in the system is maintained. High-value early adopters stay engaged rather than switching to competitors.
Purchase Decision	Studio photography on white backgrounds. No social context. No real-world styling. Customers leave Nike.com to find proof on TikTok.	Community-generated styling content surfaces automatically alongside the product page inside the app at the exact moment of decision.	Social proof at point of purchase increases conversion. Customers buy with confidence rather than abandoning to verify elsewhere first.
Post-Purchase Loop	Transaction ends the relationship. No community, no content, no mechanism to come back until the next campaign email.	Every purchase generates a styling signal. Members post content, which feeds the next drop's discovery algorithm and waitlist model.	Customers become active participants in the next drop cycle. Post-purchase engagement builds the data asset that makes each drop smarter.

Velocity's CX advantage is not a new app feature. It is a better experience at every stage of the journey that Gen Z already takes, one that Nike currently fails at each step.

Two production-ready AI technologies deliver Velocity's CX improvements, both built on infrastructure Nike already owns



AI-MANAGED DROP ALLOCATION

Replaces the SNKRS lottery with merit-based access that filters bots and rewards real fans

The CX problem it solves

The SNKRS lottery damages loyalty among your most valuable early adopters. Losing to a bot doesn't feel like bad luck, it feels like Nike doesn't care. Persona B (Jalen) in your research has already disengaged because of this.

How it improves the experience

A reinforcement learning model scores waitlist priority by engagement authenticity rather than randomness. Bot patterns are filtered. Community members are rewarded. Scarcity is maintained but fairness is restored.

What the customer feels

Even customers who don't get the product feel the system is fair. That perception keeps them coming back for the next drop rather than switching to a competitor.

Proven by: StockX | Supreme | Adidas CONFIRMED

AI COMMUNITY COMMERCE

Surfaces real community styling at the point of purchase: converting social proof into sales inside Nike's own app

The CX problem it solves

Gen Z leaves Nike.com to find social proof on TikTok before buying. Once there, they might buy from TikTok Shop instead. Nike loses the conversion and the data. Brand photography alone does not convert this customer.

How it improves the experience

A recommender system surfaces user-generated styling content from real Velocity members directly alongside the product page inside the app, social proof at the moment of highest purchase intent.

What the customer feels

Buying feels culturally validated, not transactional. Every styling post feeds the next drop's discovery algorithm, making the CX smarter with each cycle.

Proven by: Pinterest Visual Shopping | TikTok Shop | ASOS Style Feed

Both technologies run on Nike's existing 160M member data asset. No new infrastructure required, just a better job for data Nike already owns.

Challenges & Next Steps

Challenges that may limit Velocity's Success



Brand Dilution

HIGH

Fast-fashion association could undermine Nike's premium positioning if brand separation is not maintained.

Mitigation: Launch Velocity as a distinct brand identity with separate app, visual language, and AI-led storytelling — not price.

AI Bias in Trend Forecasting Level

HIGH

Models trained on Western data may miss trends from diverse and emerging communities. Incorrect signals-> costly mistakes

Mitigation: Train on geographically diverse datasets. Add human editorial review from culturally embedded curators pre-drop.

Consumer Skepticism of AI Design

MEDIUM

Gen Z may react negatively to clothing perceived as algorithm-designed rather than human-crafted.

Mitigation: Position AI as a creative amplifier, not a replacement. Lead communications with designers and collaborators.

Data Privacy & Social Ethics

HIGH

Scraping social platforms carries GDPR/CCPA legal risk and evolving API policy restrictions.

Mitigation: Partner with licensed providers (Brandwatch, Heuritech). All customer data collection MUST be consent-based.

Supply Chain Partner Readiness

MEDIUM

Nearshore partners may lack the digital infrastructure to execute 2–3 week turnarounds at launch.

Mitigation: Phase rollout with 1–2 pre-qualified nearshore partners. Invest in shared systems integration before scaling.

Sustainability

MEDIUM

Fast Cycles may increase waste and environmental footprint; fast fashion industry can face billions in unsold inventory annually due to overproduction

Mitigation: Focus on Demand based production and scoring sustainable materials

Cannibalization of Nike By You

LOW

May conflict with engagement with Nike By You and its premium made-to-order-margins

Mitigation: Clearly differentiate the two products—Nike By You is for customers who want to express themselves through hands-on design; Velocity is for customers who want AI to do the curation for them.

Velocity's risk profile is manageable — mitigatable through phased rollout, licensed data, information transparency, and brand architecture planning.

Next Steps & Roadmap



WHERE NIKE SHOULD FOCUS NEXT

1

PHASE 1 Foundation · Months 1–6

- License WGSN Instock & Trendalytics for social trend monitoring
- Activate Unified Customer Data Platform on Nike existing database
- Onboard 1–2 nearshore manufacturing partners
- Deploy Drop Performance Prediction model on historical data

2

PHASE 2 Scale · Months 7–18

- Launch AI-Managed Drop Allocation to replace SNKRS lottery
- Deploy AI Community Commerce on Velocity app product pages
- Integrate systems for real-time production scheduling
- Expand nearshore network to 4–6 partners as needed with growth

3

PHASE 3 Innovate · Months 18–36

- Pilot Generative AI Design Studio (diffusion models on Nike archive)
- Full drop cycle live on social-native channels
- Explore social-native commerce integrations (TikTok Shop, IG native)
- Build brand equity metrics dashboard for Velocity vs. Nike mainline

Expected Outcomes & ROI

\$50B+

Gen Z athleisure market

20–30%

Inventory reduction target

30 days

Design-to-shelf goal

Next Step in Final Presentation

01

Validate with Nike Stakeholders

Share Velocity's AI roadmap with Nike's internal digital and supply chain teams to stress-test assumptions and refine the build-vs-buy decision for each technology.

02

Prototype the Drop Experience

Create a mockup of the Velocity app's drop discovery and AI-managed allocation flow to present CX improvements in client-ready format.

03

Finalize Risk Mitigation Plans

Detail the legal compliance framework, brand architecture guardrails, and supplier qualification criteria needed before Phase 1 launch.

Fast fashion is a speed problem. Nike has always had the brand trust and the data to solve it. Velocity is the decision to finally do so.

Appendix

Appendix: References



- **NIKE, Inc.** (2025). Form 10-K for the fiscal year ended May 31, 2025. Retrieved from <https://investors.nike.com>
- **Statista** (2025). Nike: Companies & Products Statista Dossier (Study ID 12870). https://www.statista.com/study/12870/nike-statista-dossier/?srsltid=AfmBOooRbC9v1fxNW8dxXA80N_3-DK0YxlddDQjdAnkHrVOLyMGCP7Dz
- **BCG.** (2025). *How Gen Z and Gen Alpha Are Rewiring Fashion: Global Spending Power and Consumer Behavior Report*. Retrieved from <https://www.bcg.com/press/29october2025-gen-z-gen-alpha-drive-fashion-spending>
- **EcommerceFastlane.** (2025). *TikTok Shop GMV and Social Commerce Growth Analysis*. Retrieved from <https://ecommercefastlane.com/boost-your-tiktok-shop-sales-with-influencer-marketing/>
- **PwC.** (2025). *2025 Global Consumer Insights Survey: Buy Now Pay Later and Gen Z Financial Trends*. Retrieved from <https://www.pwc.com/gx/en/industries/consumer-markets/consumer-insights-survey.html>
- **McKinsey & Company.** (2023). *Generative AI: Unlocking the future of fashion*. McKinsey Global Institute. Retrieved from <https://www.mckinsey.com/industries/retail/our-insights/generative-ai-unlocking-the-future-of-fashion>
- **Inditex.** (2024). *Annual Report 2024: Technology and supply chain innovation*. Retrieved from <https://www.inditex.com/investors/investors-relations/annual-reports>
- **o9 Solutions.** (2025). *AI-powered demand sensing and supply chain orchestration: Enterprise case studies*. Retrieved from <https://o9solutions.com/resources>
- **Kinaxis.** (2025). *Concurrent planning in apparel and retail: Agility through AI*. Retrieved from <https://www.kinaxis.com/en/resources>
- **WGSN.** (2025). *Fashion trend forecasting methodology and retail deployment*. Retrieved from <https://www.wgsn.com/en/resources>
- **Trendalytics.** (2025). *Predictive trend analytics: How fashion brands reduce overproduction*. Retrieved from <https://www.trendalytics.co/>
- **Business of Fashion & McKinsey.** (2025). *Fashion industry 2025: AI and sustainability trends*. Retrieved from <https://www.mckinsey.com/industries/retail/our-insights/the-fashion-industry-faces-a-world-in-flux>